REFERENCES & EVIDENCE

References & Research Report

Source register, evidence boundaries and method notes

REF-06

AAKAASH ADHINAYAK

VAYU NEXA AI | Alchemy Architects | SIH 2026

Prepared 20 October 2026 | Design assessment, not a certification

1. Purpose and scope

This report records the public sources used to inform the AAKAASH ADHINAYAK technical, feasibility, viability, impact and revenue-model documents. It distinguishes the project team supplied material from authoritative external documentation. Sources support architecture and method choices; they do not prove that the deployed application has implemented the features described.

Evidence note: The public URL resolves to a page titled "AAKAASH ADHINAYAK - Verified Intelligence for Adaptive Resource Planning", but the available research interface did not expose navigable page text, runtime interactions, source code, API traces or test results. Therefore this report distinguishes observed deployment metadata and the team-supplied blueprint from proposed design and unverified implementation. It is not a source-code audit, security assessment, or certification of the live system.

2. Key sources and what they support

ID	Source	Use in the project
R1	AAKAASH ADHINAYAK URL	Project-specific deployment identity. Live features not independently extracted.
R2	Google AI Studio deployment guide	Deployment path and custom URL context; not proof of current app architecture.
R3-R4	AviationWeather.gov API/product docs	Weather connector design, endpoint limits, source freshness and server-side proxy requirement.
R5	NASA PCoE dataset repository	Candidate predictive-maintenance datasets; exact archive and terms must be verified.
R6	Google OR-Tools CP-SAT docs	Constraint solver candidate and careful handling of solver statuses.
R7	NIST AI RMF 1.0	AI validity, reliability, safety, security, transparency, explainability and risk management.
R8-R9	NIST SSDF and OWASP API Top 10	Secure lifecycle, access control and API risk checklist.
R10	Firebase Firestore security rules	Role-based data security and rules testing if Firebase is configured.
R11, R15	W3C WCAG 2.2 materials	Accessibility checklist for 2D/3D interface.
R12-R13	Google Cloud Run pricing and billing	Cost model inputs; actual rates depend on region and consumption.
R14	UN SDG 9	Thematic reference for innovation and resilient infrastructure; not certification of project impact.

3. Research interpretation

- API documentation confirms that a data product exists and explains its access pattern; it does not prove successful integration inside AAKAASH ADHINAYAK.
- A listed dataset is not considered integrated until the exact file is available, its source/licence is reviewed, fields and units are understood, and a reproducible evaluation is completed.
- Solver documentation explains status semantics; the application must still demonstrate its own solver integration and independent validator.
- Security and accessibility standards are engineering references, not evidence of compliance. Compliance requires tests, review and documented results.
- Cloud provider documentation supports a cost-estimation method, not a guaranteed free deployment or a specific monthly bill.

4. Research limitations

The public URL exposed a page title in the web retrieval environment, but no extractable page body or testable source code. The 736-page team-supplied conversation PDF supplies detailed intended requirements and previous technical recommendations; it does not substitute for reviewing source code, dataset files, cloud settings or runtime logs. The current reports are accordingly design/desk-review documents pending implementation verification.

5. Full source register

References and evidence register

Sources were checked on 20 October 2026. The app URL and the team-provided conversation extract are project-specific evidence; external standards and datasets are listed for design decisions. Dataset terms and access should be re-checked at implementation time.

[R1] AAKAASH ADHINAYAK deployed page
https://aakaash-adhinayak.ai.studio/
Public app URL supplied by the project team. Page title was visible through the web retrieval layer; application body and code were not inspectable in this run.

[R2] Google AI Studio deployment guide
https://ai.google.dev/gemini-api/docs/aistudio-deploying
Official deployment guidance for full-stack applications, Cloud Run deployment and custom ai.studio URLs.

[R3] AviationWeather.gov Data API
https://aviationweather.gov/data/api/
Official public aviation weather API documentation covering METAR, TAF and other products, query limits, rate limiting, CORS and cache files.

[R4] AviationWeather.gov data product information
https://aviationweather.gov/help/data/
Official definitions and fields for aviation weather products such as METAR.

[R5] NASA Prognostics Center of Excellence Data Set Repository
https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/
Public prognostics dataset catalogue; entries include bearing, battery and turbofan degradation simulation datasets. Individual files and terms must be checked before use.

[R6] Google OR-Tools CP-SAT Solver
https://developers.google.com/optimization/cp/cp_solver
Official optimization documentation describing constraint modelling and solver statuses such as OPTIMAL, FEASIBLE, INFEASIBLE, MODEL_INVALID and UNKNOWN.

[R7] NIST AI Risk Management Framework 1.0
https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf
Risk-management reference for trustworthy, valid, reliable, safe, secure, accountable, transparent, explainable and privacy-aware AI.

[R8] NIST Secure Software Development Framework, SP 800-218
https://csrc.nist.gov/pubs/sp/800/218/final
Secure software lifecycle practices to reduce vulnerabilities and improve software assurance.

[R9] OWASP API Security Top 10 - 2023
https://api-security.owasp.org/editions/2023/en/0x11-t10/
Common API risk categories, including object-level authorisation and authentication failures.

[R10] Firebase Cloud Firestore Security Rules
https://firebase.google.com/docs/firestore/security/get-started
Official guidance on role-based access patterns and testing rules; server libraries require IAM protection independently of client rules.

[R11] W3C Web Content Accessibility Guidelines (WCAG) 2.2
https://www.w3.org/TR/WCAG22/
Accessibility recommendation covering users with a range of disabilities and interaction needs.

[R12] Google Cloud Run pricing
https://cloud.google.com/run/pricing
Official usage-based pricing reference; actual costs depend on region, traffic, resource allocation, networking and free-tier eligibility.

[R13] Google Cloud Run billing settings
https://docs.cloud.google.com/run/docs/configuring/billing-settings
Official description of request-based versus instance-based billing.

[R14] UN Sustainable Development Goal 9
https://india.un.org/en/sdgs/9
UN framing for resilient infrastructure, sustainable industrialisation and innovation; alignment should be stated as thematic, not certified impact.

[R15] W3C WCAG 2.2 overview
https://www.w3.org/WAI/news/2023-10-05/wcag22rec/
W3C announcement that WCAG 2.2 became a web standard recommendation.

Evidence discipline: links establish documentation and potential resources, not that the current app has integrated them. No code bundle, request logs, dataset files, security-rule export, model artifacts or test results were supplied for independent verification.

6. Project material reviewed

Team-supplied conversation export: “ChatGPT-SIH 2026 Participation Guide-20261020-1424.pdf”, especially pages 713-729. These pages set out the Evidence-to-Decision Intelligence differentiator, data strategy, preferred architecture, security/accessibility direction, 35 challenge areas, domain model and acceptance tests.

Application URL: https://aakaash-adhinayak.ai.studio/. Accessible identity metadata was reviewed on 20 October 2026. No source bundle or credentials were requested or used.

How to upgrade this report to an implementation audit: provide the exported app source or linked Git repository, dependency/build output, route screenshots, sample data with README/licence, server logs with secrets removed, Firestore security rules if used, and test output. A runtime audit can then separate working, partial, broken, planned and blocked features with evidence.